$$\bar{v}\cos\theta\triangle t = d \ . \qquad (7)$$

$$\triangle p_{\mathrm{du}}$$

$$f = \frac{\triangle p_{\mathrm{du}}}{\triangle t} = \frac{p_{\mathrm{d}} - p_{\mathrm{u}}}{d/(\bar{v}\cos\theta)} \approx \frac{(p_{\mathrm{d}} - p_{\mathrm{u}})\, c\cos\theta}{d} \ ; \qquad (8)$$

$$f \approx 1.04 \times 10^{-9}\ \mathrm{N} \ . \qquad (9)$$

（III）

$$\triangle t = \frac{d}{\bar{v}\cos\theta} \approx \frac{d}{c\cos\theta} \approx 2.07 \times 10^{-11}\ \mathrm{s} = 0.0207\ \mathrm{ns} \ , \qquad (10)$$

τ =2ns0

$$F \approx (p_{\mathrm{d}} - p_{\mathrm{u}})\, j \ ; \qquad (11)$$

$$F = 0.107\ \mathrm{N} \ . \qquad (12)$$

（4）

$$E = \sqrt{p^2c^2 + m^2c^4} \ , \qquad (13)$$

$$E_{\mathrm{d}} = \sqrt{p_{\mathrm{d}}^2c^2 + m^2c^4} = 63.0\ \mathrm{MeV} \ , \ E_{\mathrm{u}} = \sqrt{p_{\mathrm{u}}^2c^2 + m^2c^4} = 22.8\ \mathrm{MeV} \ . \qquad (14)$$

$$Q = (E_{\mathrm{d}} - E_{\mathrm{u}})\, j\tau \ ; \qquad (15)$$

$$\cdot\ Q = 8.04 \times 10^{-2}\ \mathrm{J} \ . \qquad (16)$$